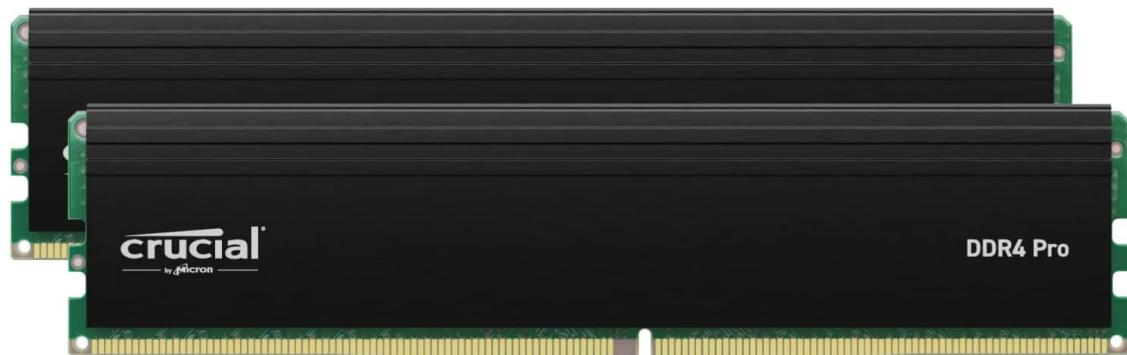
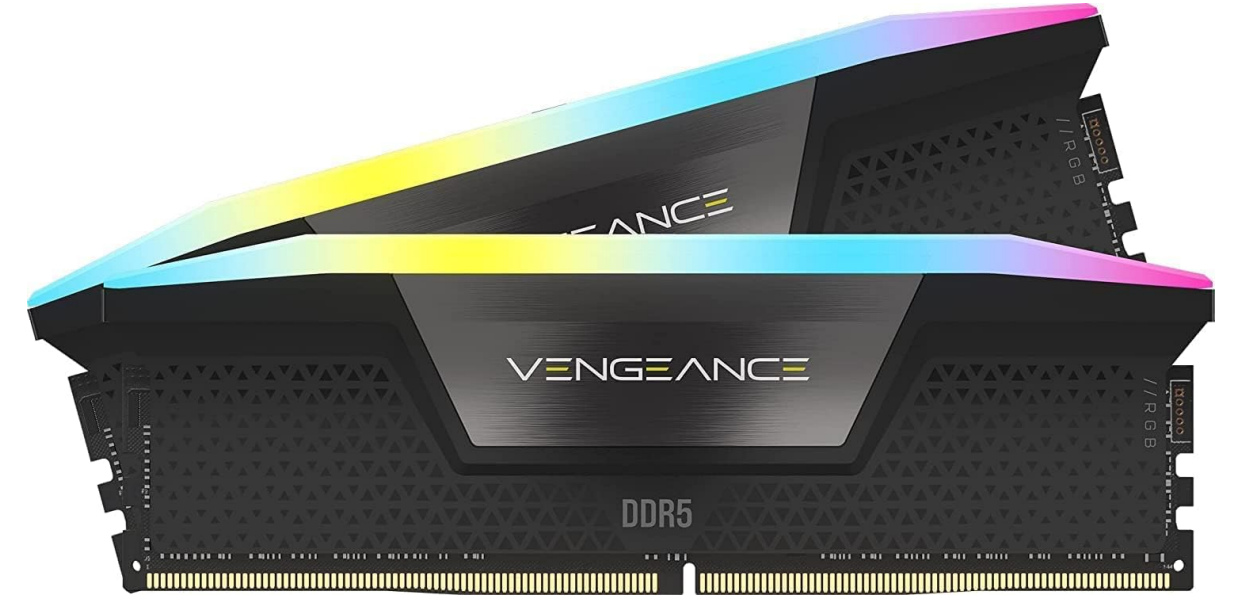


Choosing Memory

DDR3 or DDR4 or DDR5?

?



8GB, 16GB
or 32GB?

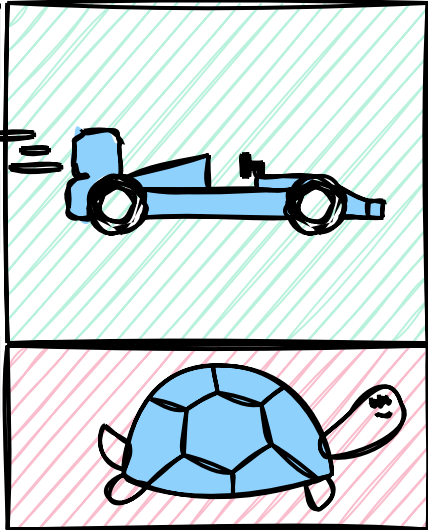
Quick intro to RAM

Here's how fast RAM is



Storage Type	Bandwidth (Approx)	Latency (Approx)
L1 Cache	1 TB/s	1ns = 80x faster than RAM
L2 Cache	1 TB/s	4ns = 20x faster than RAM
L3 Cache	400 GB/s	40ns = 10x faster than RAM
DDR 5 RAM	32-64 GB/s	80ns
M.2	12 GB/s	2800ns = 35x slower than RAM
SATA SSD/HDD	0.75 GB/s	1,000,000ns = 1200x slower than RAM

how fast you get a response



More RAM = less loading from disk, faster response (imagine 100 browser tabs open)

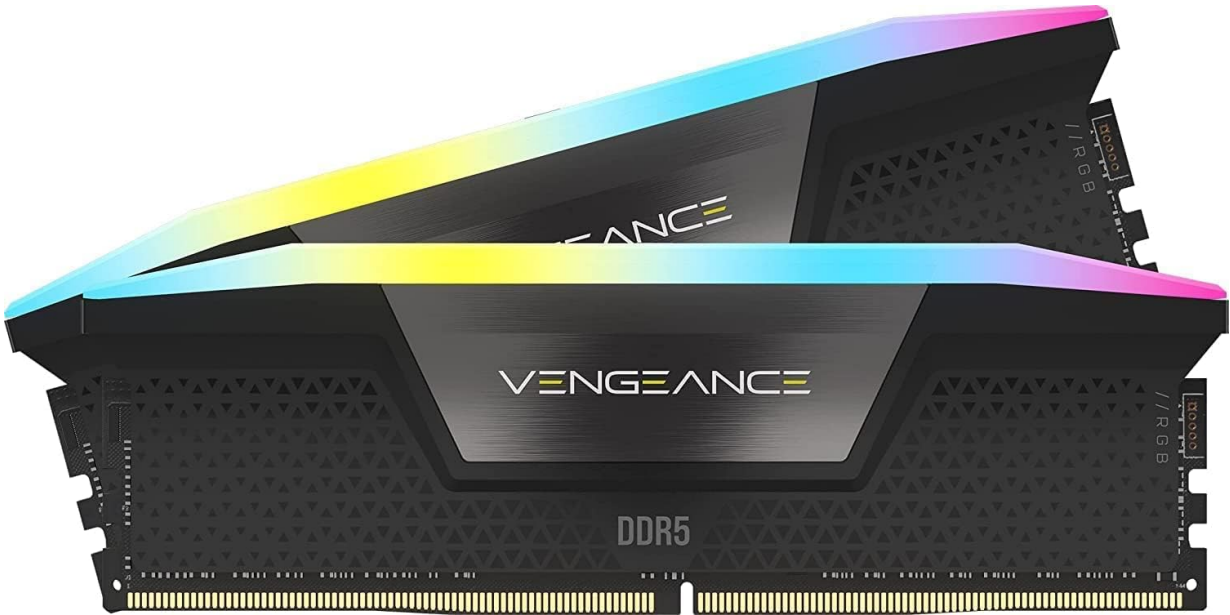
SO-DIMM



Small Outline DIMM
Laptop Memory

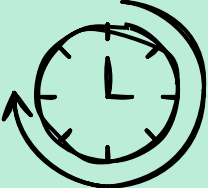
DDR

Desktop Memory



DDR = Double Data Rate
Evolution of SDR (Single Data Rate)

Generations of DDR



DDR1 (1998)

DDR2 (2003)

DDR3 (2007)

DDR4 (2014)

DDR5 (2020) Current

DDR6 (2026?)

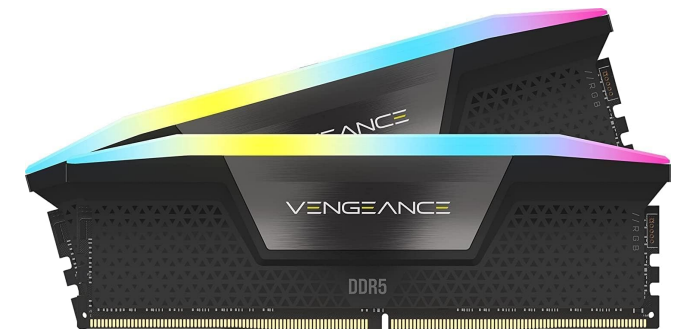
Types of RAM

All DDR of a generation is not equal...
There are speed ratings
within each generation

Speed	Industry Name	Data Rate (Megabytes)	Transfers / Second (Hz)
DDR5-8000	PC5-64000	64000 MB/s	8000 MT/s
DDR5-7600	PC5-60800	60800 MB/s	7600 MT/s
DDR5-7200	PC5-57600	57600 MB/s	7200 MT/s
DDR5-6800	PC5-54400	54400 MB/s	6800 MT/s
DDR5-6400	PC5-51200	51200 MB/s	6400 MT/s
DDR5-6200	PC5-49600	49600 MB/s	6200 MT/s
DDR5-6000	PC5-48000	48000 MB/s	6000 MT/s
DDR5-5600	PC5-44800	44800 MB/s	5600 MT/s
DDR5-5200	PC5-41600	41600 MB/s	5200 MT/s
DDR5-4800	PC5-38400	38400 MB/s	4800 MT/s
DDR5-4400	PC5-35200	35200 MB/s	4400 MT/s
DDR5-4000	PC5-32000	32000 MB/s	4000 MT/s
DDR4-4400	PC4-35200	35200 MB/s	4400 MT/s
DDR4-4000	PC4-32000	32000 MB/s	4000 MT/s
DDR4-3600	PC4-28800	28800 MB/s	3600 MT/s
DDR4-3200	PC4-25600	25600 MB/s	3200 MT/s
DDR4-3000	PC4-24000	24000 MB/s	3000 MT/s
DDR4-2933	PC4-23400	23400 MB/s	2933 MT/s
DDR4-2666	PC4-21300	21300 MB/s	2666 MT/s
DDR4-2400	PC4-19200	19200 MB/s	2400 MT/s
DDR3-1600	PC3-12800	12800 MB/s	1600 MT/s
DDR3-1333	PC3-10600	10667 MB/s	1333 MT/s
DDR3-1066	PC3-8500	8533 MB/s	1066 MT/s
DDR3-800	PC3-6400	6400 MB/s	800 MT/s

The Memory Market

Big names....



Very diverse and open (budget brands appear)

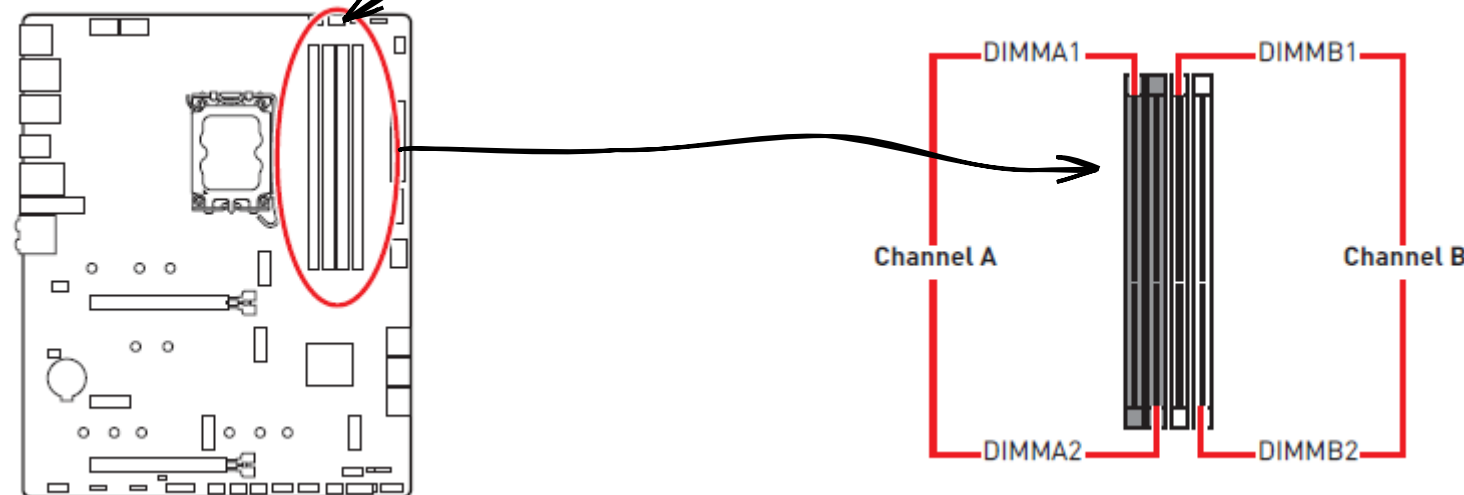


Don't discount budget memory with good reviews, it's easy to test, review and feedback on

Why Memory Pairs?



Typical ATX board has 4 slots, 2 channels

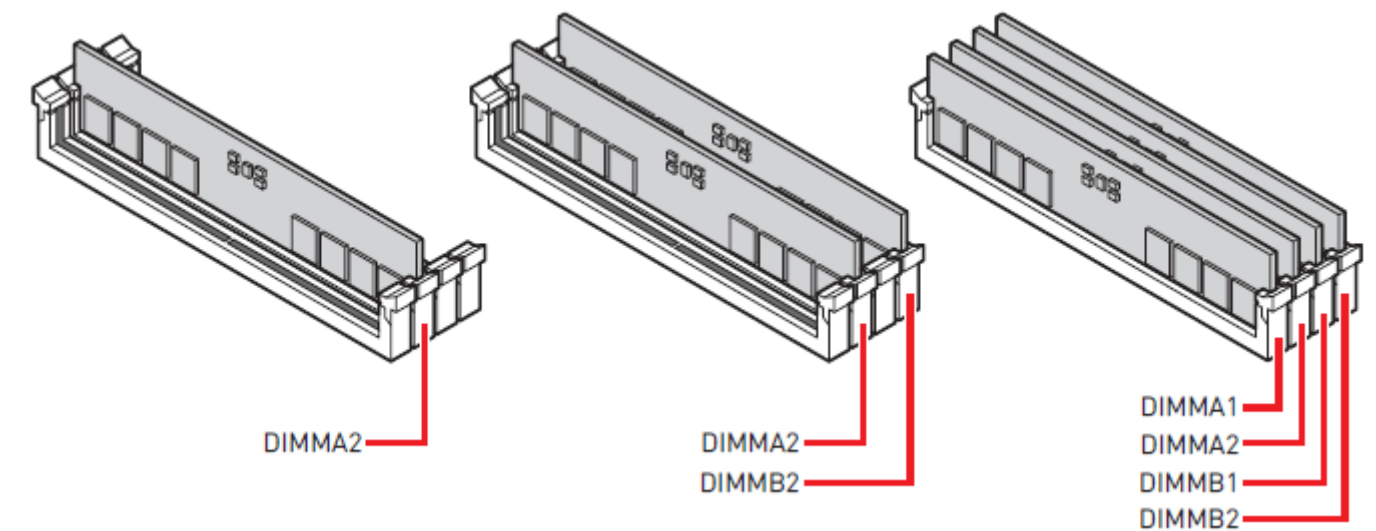


Matching pairs are necessary

Non matching pairs won't get fully utilized, (i.e. 8+16 will only give you 16)

Typical mobo slot installation recommendations

Slot 1 (Channel 1)	Slot 2 (Channel 1)	Slot 3 (Channel 2)	Slot 4 (Channel 2)	Total
	8 GB			8 GB
	8 GB		8 GB	16 GB
8 GB	8 GB	8 GB	8 GB	32 GB
	16 GB		16 GB	32 GB
	32 GB			32 GB
32 GB	32 GB	32 GB	32 GB	128 GB



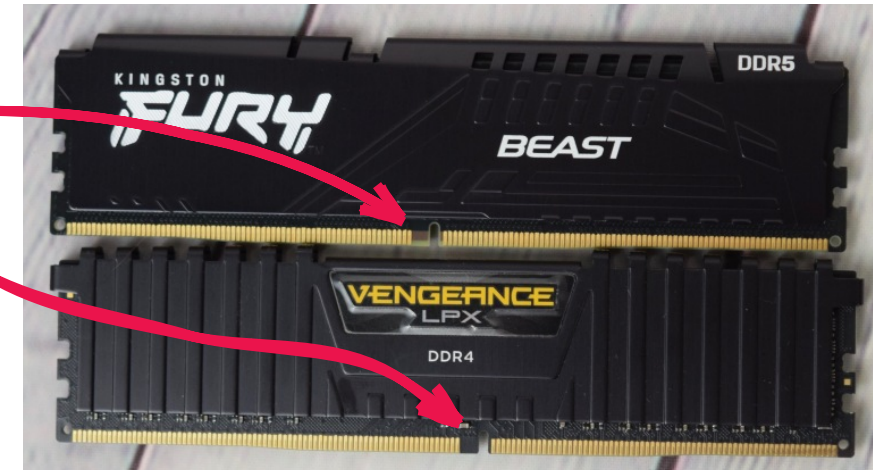
Other notes on memory

DDR generations are NOT interchangeable

To use DDR4 you need a DDR4 motherboard

To use DDR5 you need a DDR5 motherboard

And so on for other types of DDR...



DDR5

DDR4

Intel XMP and AMD EXPO

Safely overclockable memory

Overclocks are easily applied by switching profiles in the BIOS

Memory needs to be certified and motherboard needs support



LEDs

LED memory just works

Controlled from the motherboard

No cables

ECC memory is for servers

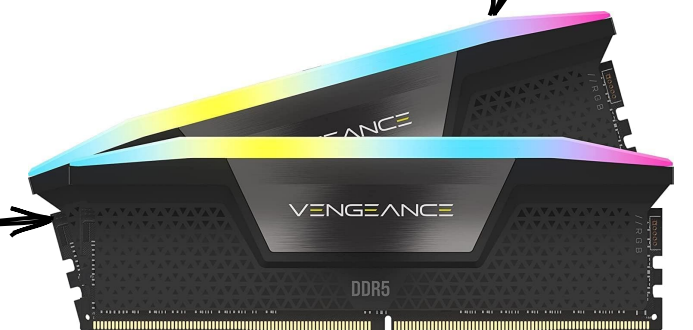
ECC = Error Correction Code

Used in servers

Overkill for most desktop users

Slower than regular RAM (checking for errors takes time)

Budget DDR (No heat spreader)



Heat Spreaders

Common these days, DDR can get warm

Decoding a Memory Listing



G.Skill Trident Z5 NEO RGB 32GB 6000MHz CL32 DDR5 Memory

- Capacity: 32GB (2x16GB)
- Speed: 6000MHz
- Latency Timing: CL32
- Form Factor: DDR5 UDIMM
- Warranty: Limited Lifetime

Cas Latency

Remember latency? How quick you get an answer?

Two listings, same size and speed, different CL

CL32 memory will be faster than CL36 (and more expensive because of that)



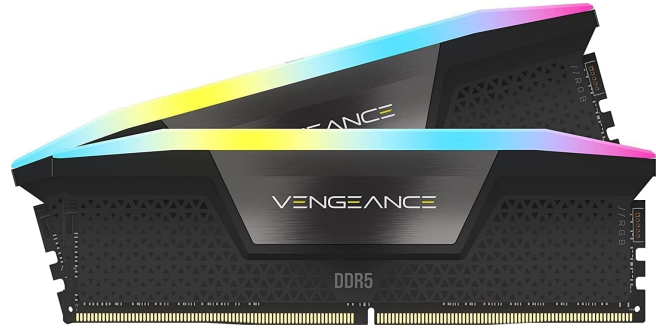
Corsair Vengeance RGB 32GB DDR5 6000MHz CL36 Desktop Memory

- Capacity: 32GB (2x16GB)
- Speed: 6000MHz
- Latency Timing: CL36
- Form Factor: DDR5 DIMM
- Warranty: Limited Lifetime

Speed	Industry Name	Data Rate (Megabytes)	Transfers / Second (Megabits)
DDR5-8000	PC5-64000	64000 MB/s	8000 Mb/s
DDR5-7600	PC5-60800	60800 MB/s	7600 Mb/s
DDR5-7200	PC5-57600	57600 MB/s	7200 Mb/s
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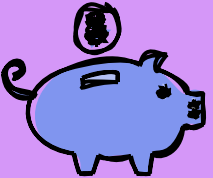


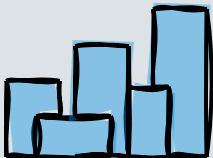
Memory Recommendations



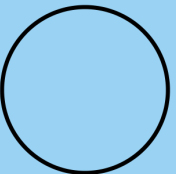
Budget/Casual

Min 4GB (Windows 11 Requirements)
Recommended 8GB



WFH Pro 

Min 8GB
Recommended 16GB

Gamer 

Min 16GB
Recommended 32GB (Budget permitting)

**Creative Pro
/ Engineer** 

Min 32GB
Recommended 32GB+